

QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 06

Climate - Regional, Site & Measurement

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	6.0 - Climate (13 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Method	The 13-slide deck is mapped across two major families: regional/site climate and field measurement of temperature, humidity and air movement. Site-climate questions are included even when historical papers placed them under the opposite printed section, consistent with the current Section A lecture sequence. Large-site design-process and urban-climate questions are marked Partial when the deck supplies only baseline data/factors.

RESULT

Lecture 06 is a very high-return site-climate and measurement lecture. Site/microclimate factors recur in every analyzed year, while Stevenson Screen / temperature-humidity-airflow measurement appears repeatedly from 2017-2024. The deck also directly supports the 2020/2021 RH calculation family. Because it combines definitions, factor hierarchies, instruments and a formula, this lecture links to more than twenty historical subquestions.

Prepared as part of the cumulative lecture-by-lecture question bank analysis.

1. Executive Summary

Lecture 06 organizes climate by scale (global, regional, site), breaks regional climate into primary/secondary/tertiary/local factors, lists direct site-climate modifiers, and then explains why and how designers measure climate. The measurement block covers long-term averages/extremes, thermometer/Stevenson Screen, humidity definitions and RH relation, and anemometer-based airflow measurement.

13

Slides reviewed

23

Question links

15

Exact matches

4

Direct matches

4

Partial matches

265

Mapped marks

Lecture structure and exam value

Slides	Lecture block	Historical signals	Exam value
2	Global / regional / site climate types	2017 Q3(a); 2022 Q4(a); 2021 urban-vs-macro partial	Useful comparison framework; definitions should be expanded from Slide 3-4.
3	Regional climate factor hierarchy	2016 Q4(a); 2017 Q7(b), Q4(c); 2020 Q6(b); 2024 Q3(a support	High-value factor slide: primary, secondary, tertiary, local/minor.
4	Site climate factors	2016, 2017, 2018, 2020, 2021, 2022, 2024	One of the most repeated factor lists in the entire question bank.
5-6	Why designers need climatic data + long-term averages/extremes	2020/2022/2023 site-design questions partial; 2024 climatic measurement support	Important for method/process answers but not a full site-design algorithm.
7-9	Temperature measurement + Stevenson Screen	2017, 2018, 2021, 2022, 2024	Repeated instrument/sketch family. Draw the screen and explain white louvers, double roof and raised stand.
10-11	Humidity: AH, RH, formula and significance	2020 Q2(b); 2021 Q6(d)	Direct formula/numerical support.
12	Air movement measurement	2024 Q7(c)	Direct anemometer + unit support.
13	Reference	No direct question	Navigation only.

Priority note

If you prepare only three blocks, choose Slides 3-4 (site/regional factors), Slides 7-9 (Stevenson Screen), and Slides 10-12 (humidity + airflow measurement).

2. Detailed Year-wise Question Mapping

Original paper references are preserved. Match labels mean: Exact = the slide block substantially answers the task; Direct = the requested concept is clearly taught but some expansion/sketch/application may still be needed; Partial = the deck only supports part of the requested answer.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Q4(a)	25	1	Identify the local factors responsible for developing a site or micro climate. Which is different from regional or macro climate of the same area?	Exact	2-4	Slides 2-4 provide the climate-scale distinction plus regional and direct site-climate factors.
2017	Q3(a)	15	2	Differentiate Site Climate from Regional Climate.	Exact	2-4	The deck explicitly distinguishes scale and factor structure for regional vs site climate.
2017	Q4(a)	10	2	What is microclimate? What do you know about the micro climate factors?	Direct	3-4	The factors are direct; the word "microclimate" is not separately defined in a formal sentence.
2017	Q4(b)	15	2	What is site climate? Which factors are responsible for site climate?	Exact	2-4	Direct climate-scale definition/context plus the six site-climate factors.
2017	Q4(c)	10	2	Write short notes on Ocean current and Natural wind breaker.	Direct	3-4	Both appear as regional/site modifiers; detailed mechanisms are only briefly implied.
2017	Q7(b)	10	2	What are the tertiary factors constituting the regional climate?	Exact	3	Slide 3 lists tertiary factors explicitly.
2017	Q7(a-iii)	5	2	Short note: Stevenson Screen.	Exact	8-9	Slides 8-9 provide construction and working features.
2018	Q3(b)	8	3	Show the working mechanism of Stevenson Screen for measuring temperature.	Exact	7-9	The deck provides both instrument context and a labelled Stevenson Screen diagram.
2018	Q6(a)	20	4	What is site climate? Elaborate the factors of site climate.	Exact	2-4	Direct core Lecture 06 question.
2020	Q2(b)	12	5	Given AH = 10 g/m3 and SH = 17 g/m3 at 20 C, calculate RH.	Exact	10-11	Slide 10 gives $RH = AH/SH \times 100\%$; Slide 11 clarifies AH/RH meaning.
2020	Q3(a)	10	5	Describe the fundamental process of designing a site as a designer using site-climate knowledge in the case of a large site.	Partial	5-6	The deck explains what climatic data designers need and the importance of long-term averages/extremes, but not the full step-by-step site-design process.
2020	Q3(b)	20	5	Explain the factors that contribute to the formation of a site climate.	Exact	3-4	The lecture directly lists regional modifiers and six direct site-climate factors.

2. Detailed Year-wise Question Mapping - Continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2020	Q6(b)	15	5	Identify the local factors that are responsible for developing a site or micro climate. Which is different from regional or macro climate of the same area?	Exact	2-4	Near-repeat of the 2016 question and a direct deck match.
2021	Q3(a)	10	6	Distinguish between urban climate and macroclimate.	Partial	2-4	Macro/regional/site scale is present, but urban-climate mechanisms and deviations are not taught in this deck.
2021	Q3(c-ii)	5	6	Short note: Stevenson Screen.	Exact	8-9	Direct repeated instrument question.
2021	Q4(b)	10	6	What is the climate? Which factors are responsible for site climate?	Direct	2-4	The climate-scale framework and site factors are direct; a broad climate definition may need Lecture 01/02 wording.
2021	Q6(d)	5	7	Define AH and SH and calculate RH when AH = 10 and SH = 17 g/m3.	Direct	10-11	AH and RH are direct; saturation humidity is implied by the RH denominator but not separately defined on these slides.
2022	Q3(b)	10	8	Write down the instrument required to measure temperature data in an open field. Discuss the working mechanism of this instrument.	Exact	7-9	Thermometer + Stevenson Screen construction/working are directly shown.
2022	Q4(a)	10	8	Distinguish between site climate and regional climate.	Exact	2-4	Direct repeat of the lecture structure.
2022	Q4(b)	10	8	Describe the fundamental process of designing a site as a designer using site climate knowledge.	Partial	5-6	Data needs are covered, but not the complete design process.
2023	Q3(b)	10	10	Describe the fundamental process of designing a residential building at KUET as a designer using site climate knowledge.	Partial	5-6	Useful baseline on data and extremes; detailed site-analysis/design sequence must come from expanded site-climate material.
2024	Q3(a)	10	13	What is site climate, and what factors cause deviations in local climate?	Exact	2-4	Direct match: site climate plus local modifiers.
2024	Q7(c)	10	14	Describe the process of measuring climate parameters - temperature, humidity and air flow - from a specific site for designing a climate-responsive building.	Exact	7-12	The measurement block directly covers all three parameters, instruments and units.

3. Repeated Question Patterns and Quick Recognition

These patterns compress historical wording into the question families that matter for revision.

P	Normalized repeated question	Years seen	Slides	Why it matters
1	Site/microclimate factors and local deviation	2016, 2017, 2018, 2020, 2021, 2022, 2024 (+ 2023 process)	2-4	One of the most stable question families in the whole course.
2	Site climate vs regional/macro climate	2016, 2017, 2020, 2022	2-4	Repeated comparison wording; learn scale + modifiers.
3	Stevenson Screen / temperature measurement	2017, 2018, 2021, 2022, 2024	7-9	Repeated sketch/instrument family.
4	Climatic data for site design / large-site process	2020, 2022, 2023	5-6	Deck gives data logic but not full design algorithm; important handoff to expanded site-climate material.
5	AH / RH relation and numerical	2020, 2021	10-11	Simple formula marks; memorize units and relationship.
6	Measure temperature, humidity and airflow	2024	7-12	Recent integrated measurement question; likely useful as a modern-paper pattern.

Quick recognition rules

Question clue	Go first to	What Lecture 06 can do
"site climate factors / microclimate factors"	Slides 3-4	Exact hierarchy + direct site factor list.
"site vs regional climate"	Slides 2-4	Use scale, geographical extent and factor hierarchy.
"tertiary factors"	Slide 3	Exact list: soil, topography, natural/man-made windbreakers, water bodies, cities/industries, deserts.
"Stevenson Screen"	Slides 8-9	Draw white louvered box, double roof, raised stand; explain shade + ventilation.
"measure temperature / humidity / airflow"	Slides 7-12	Thermometer + screen; wet/dry bulb or hygrometer; anemometer.
"RH from AH/SH"	Slide 10	$RH = AH/SH \times 100\%$.
"design a large site using site climate"	Slides 5-6 then expanded notes	Only data needs/averages/extremes are here; full process is not.
"urban climate"	Do not stop here	Lecture 06 names scale but does not explain urban-climate deviation mechanisms.

4. Slide-to-Question Reverse Index

Use this page when studying directly from the lecture deck: open the slide block, then practice the linked question family immediately.

Slides	Topic	Past questions	Study note
2	Types of climate: global, regional, site	2016 Q4(a); 2017 Q3(a); 2021 Q3(a) partial; 2022 Q4(a)	Use as the opening comparison framework.
3	Regional-climate factors: primary, secondary, tertiary, local/minor	2016 Q4(a); 2017 Q4(c), Q7(b); 2020 Q6(b); 2024 Q3(a) support	High-value hierarchy slide; memorize categories and examples.
4	Site-climate factors	2016 Q4(a); 2017 Q4(a,b); 2018 Q6(a); 2020 Q3(b), Q6(b); 2021 Q4(b); 2024 Q3(a)	Probably the single most repeated slide in Lecture 06.
5-6	Why measure climate + long-term averages/extremes	2020 Q3(a); 2022 Q4(b); 2023 Q3(b); 2024 Q7(c) support	Use in method answers, but add the full site-design sequence from expanded notes.
7	Temperature instrument / Celsius / dry-bulb thermometer	2018 Q3(b); 2022 Q3(b); 2024 Q7(c)	Basic measurement intro.
8-9	Stevenson Screen construction and working	2017 Q7(a-iii); 2018 Q3(b); 2021 Q3(c-ii); 2022 Q3(b)	Must-draw sketch block.
10-11	Humidity: AH, RH, RH formula and built-environment relevance	2020 Q2(b); 2021 Q6(d); 2024 Q7(c) support	Memorize RH formula and units; note that SH definition is not explicit here.
12	Air movement measurement by anemometer, m/s	2024 Q7(c)	Direct recent integrated-measurement support.
13	Reference	No direct question	No study time needed.

5. Coverage Gaps and Diagnostic Exclusions

Lecture 06 is strong on factor lists and measurement but weaker on full design-process narratives and urban-climate mechanisms. Those gaps should be handled explicitly rather than inferred.

Gap	Affected questions	Action
Full step-by-step site / residential design process	2020 Q3(a); 2022 Q4(b); 2023 Q3(b)	Slides 5-6 explain data needs, averages and extremes only. Use expanded site-climate notes for the actual sequence from regional data to zoning/orientation/envelope.
Urban climate vs macroclimate / human intervention / urban deviation	2021 Q3(a) and later urban-climate questions	Not developed in this deck. Use dedicated site/urban-climate lecture material.
Formal definition of microclimate	2017 Q4(a)	The deck lists scale/factors but does not give a standalone textbook definition.
Saturation Humidity (SH) definition	2021 Q6(d)	RH relation is given, but SH is not separately defined on Slides 10-11. Use humidity notes for exact definition.
Detailed mechanisms of local modifiers such as ocean current / windbreakers	2017 Q4(c)	Slide 3 names them; full explanatory mechanisms require class notes/reference material.

**Mapping
discipline**

The very high historical frequency of site-climate questions does not mean every site-climate subtopic is in Lecture 06. Exact mapping is limited to the scale/factor and measurement content actually shown on these 13 slides.

6. Recommended Study Plan

Study Lecture 06 as two mini-lectures: first site/regional climate, then climate measurement. The factor slides should be memorized before the measurement instruments.

Step	Study block	Slides	Practice immediately
1	Regional vs site climate + factor hierarchy	2-4	2016 Q4(a); 2017 Q3(a), Q4(a,b), Q7(b); 2018 Q6(a); 2020 Q3(b), Q6(b); 2022 Q4(a); 2024 Q3(a)
2	Stevenson Screen	7-9	2017 Q7(a-iii); 2018 Q3(b); 2021 Q3(c-ii); 2022 Q3(b)
3	Humidity and RH formula	10-11	2020 Q2(b); 2021 Q6(d)
4	Airflow measurement	12	2024 Q7(c)
5	Climatic data needs / averages / extremes	5-6	2020 Q3(a); 2022 Q4(b); 2023 Q3(b) - then complete from expanded notes

Exam-ready minimum from this lecture

Priority	What to reproduce	Exam technique
A	Regional/site factor hierarchy	Use a four-level hierarchy table + separate six-item site-factor list.
A	Stevenson Screen sketch	Label white paint, louvers, double roof, raised stand and instruments.
A	Temperature-humidity-airflow instruments	Use a three-row table: parameter, instrument, unit/field condition.
B	RH formula	Write $RH = AH/SH \times 100\%$ and show one substitution example.

Bottom line

Lecture 06 is worth preparing thoroughly because its content is both repeated and concrete. Slides 3-4, 8-10 and 12 should be instantly reproducible under exam conditions.